TOP-Q — Day 6 worked notes

Monte Carlo study 2: does pinning protect the phase an interferometer sees?

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All results are **numerical simulations** [Simulated]. Arrays T and D ($M = 81$), centre-guide launch, device length $L = 10$ coupling lengths $= 5\pi/\kappa_2$ unless stated. 2000 simulated chips per point = 1000 independent arm pairs. Units: σ_β in κ_2, σ_κ dimensionless.

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1 The question, made measurable

An interferometer never measures an absolute phase. It measures the **difference** between two arms. So the Day-6 experiment (Fig. 1) is: put two arrays of the *same design* on one chip, give each its own random local errors, and compute

$$\Delta\varphi = \delta\varphi_A - \delta\varphi_B,$$

where $\delta\varphi$ is each arm's phase error at the output (both definitions from Day 5: modal and output-guide). A chip-wide error (e.g. the whole wafer slightly thicker) shifts both arms equally and **cancels exactly** in $\Delta\varphi$ [Derived].

1.1 Two numbers we report

- **Circular standard deviation** of $\Delta\varphi$. Phases wrap around at 2π, so the ordinary std is misleading for large errors. We use $R = |\langle e^{i\Delta\varphi}\rangle|$ and $\sigma_{\text{circ}} = \sqrt{-2\ln R}$, which equals the ordinary std for small Gaussian errors [Verified-by-test].
- R **itself**, which has a direct meaning: if the arms of a 50:50 interferometer have a random phase difference $\Delta\varphi$, the *average* fringe contrast left over is R [Derived: average of $\cos(\theta + \Delta\varphi)$]. $R = 1$: perfect; $R = 0$: no interference left on average.

First-order prediction for the modal phase: $\sigma(\Delta\varphi) \approx \sqrt{2} \cdot$ (Day-3 slope) $\cdot\, \sigma \cdot L$ (two independent arms, error grows with length).

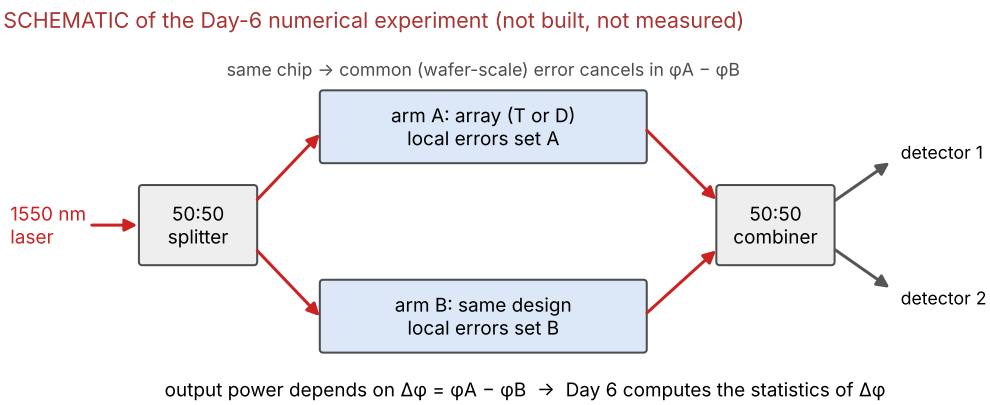


Figure 1: **Schematic** of the numerical experiment (drawn by us; not built, not measured).

2 Results

2.1 Gap errors (chiral): phase protected

T: $\Delta\varphi = 0$ *exactly* (both definitions) for every σ_κ up to 0.17; at $\sigma_\kappa = 0.3$ the output-guide phase flips to π in 0.2% of arm pairs (the Day-5 sign flip when the zero-mode weight falls below 1/2), giving $\sigma_{\text{circ}} = 0.09$ rad and $R = 0.996$. D: $\sigma_{\text{circ}} = 0.35$ rad at $\sigma_\kappa = 0.03$ and 1.04 rad at 0.095, matching first order (0.34, 1.07). Contrast at $\sigma_\kappa = 0.095$: **T 1.00, D 0.58**. [Simulated]

2.2 Width errors (not chiral): no protection

T and D are the same within statistics: $\sigma_{\text{circ}} = 0.17$ / 0.17 rad at $\sigma_\beta = 0.0095$; 0.53 / 0.53 at 0.03; contrast at 0.095: 0.23 / 0.24. First order predicts 0.171 and 0.540 for both. [Simulated]

2.3 Realistic mix (width model, $\sigma_\beta = \sigma_\kappa$)

T's phase error is about 0.66 of D's (0.54 vs 0.83 rad at $\sigma = 0.03$), exactly the Day-4 eigenvalue ratio: the phase inherits it through $\delta\varphi \approx \delta b\, L$. As on Day 4, this ratio depends on the unknown $\sigma_\beta/\sigma_\kappa$ of real waveguides.

2.4 Length: errors grow in proportion

Fig. 2d: every unprotected phase error grows linearly with L; T under gap errors stays at 0 for all lengths up to 20 coupling lengths.

2.5 A tolerance number

Requiring $\sigma_{\text{circ}} < \pi/4$ at 10 coupling lengths (a loose target) allows on-site disorder only up to about $\sigma_\beta \approx 0.045\,\kappa_2$ *for both arrays*. In words: the random spread of each guide's propagation constant must be below about 5% of the strong coupling. Converting this to nanometres of width needs $\partial\beta/\partial w$ (mode solver; open question Q-ratio).

2.6 The surprise: T has MORE power imbalance under gap errors

Fig. 2f. Spread of the arm-to-arm difference in output power:

	T, gap errors	D, gap errors	T, width errors	D, width errors
$\sigma = 0.003$	0.005	0.001	0.0001	0.002
$\sigma = 0.03$	0.043	0.011	0.009	0.016

Under gap errors, T's output power varies **4–5× more** than D's, and it grows in proportion to σ (first order). Reason [Derived (qualitative)]: the pinned zero mode keeps $b = 0$ but its *shape* changes (Day 4), so its weight w_0 on the input/output guide changes in proportion to the coupling error; the output power $\approx w_0^2 + \ldots$ follows. Under width errors, T's power changes only at second order (its phase moves, its shape barely does).

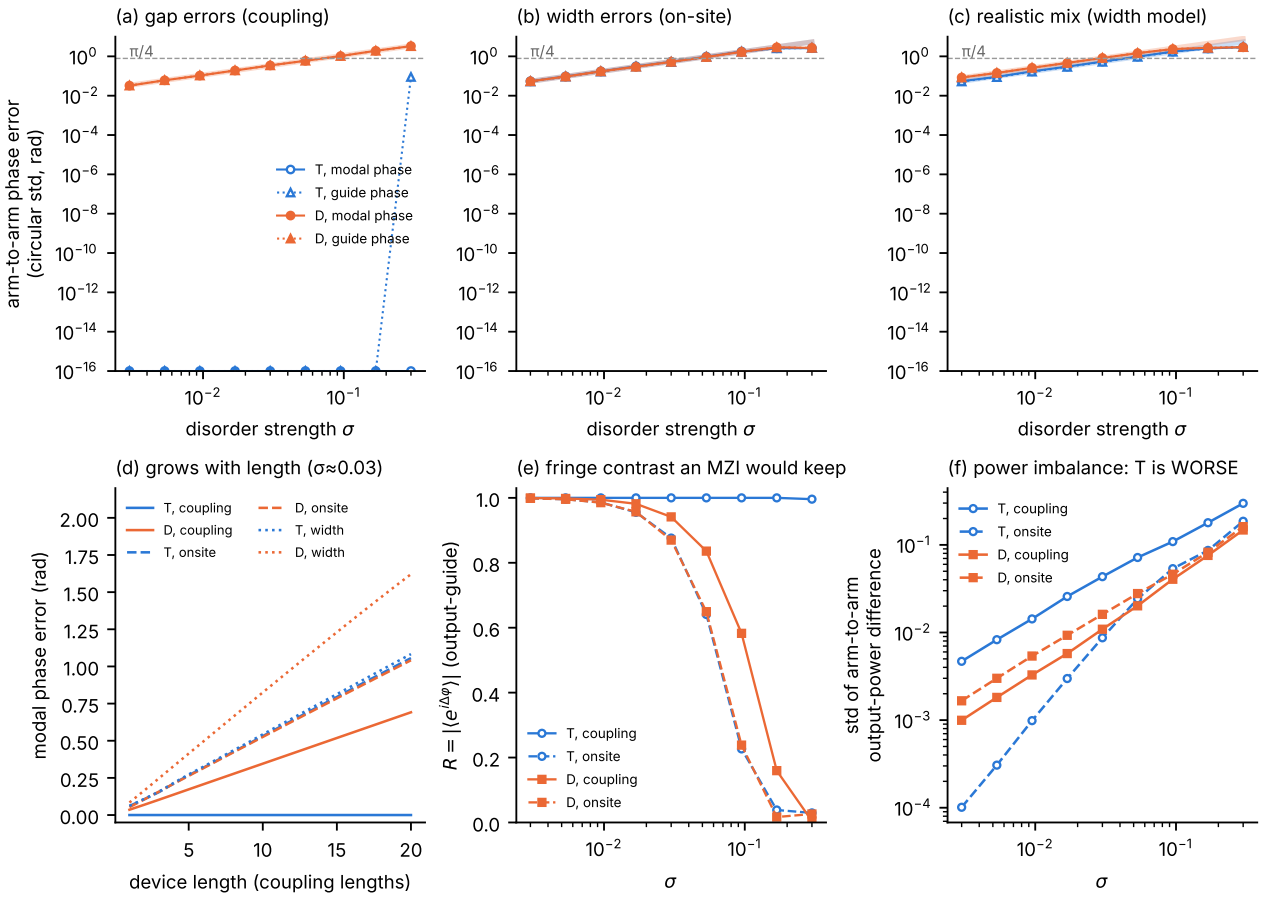


Figure 2: **Simulation**, `scripts/day6_mc_phase.py` → `data/day6_mc_phase.npz`. (a)–(c) Arm-to-arm phase error vs. disorder strength (circles/squares: modal phase; triangles: output-guide phase; thick faint lines: first-order prediction). (d) Growth with device length at $\sigma \approx 0.03$. (e) Fringe contrast R an interferometer would keep. (f) Spread of the arm-to-arm *power* difference at the output.

The pattern across Days 4–6. Topological protection *moves* gap errors from the phase into the amplitude: it does not remove them. Loss behaves the same way (Day 5). For an interferometer, an amplitude imbalance between arms also lowers the contrast — Days 8–10 will put both effects into one number.

3 What Day 6 answers

Question	Answer (simulation, this model)
Does pinning protect the arm-to-arm phase?	**Yes, exactly, for gap (coupling) errors** — both phase definitions — until the zero mode loses half its weight on the port (rare below $\sigma_\kappa \approx 0.3$).
For width errors?	**No.** Identical to an ordinary defect of the same confinement.
For a real mix?	Partly; the benefit depends on $\sigma_\beta/\sigma_\kappa$ (unknown).
For free?	**No.** Under gap errors T's output power varies more than D's.
Chip-wide errors?	Cancel in any two-arm device, topological or not.

4 Hardware: how arm-to-arm phase errors are measured on real chips

Not done by us. The images in our chat (from published papers) show the standard methods:

- **Unbalanced on-chip MZIs:** the two arms have different lengths, so the transmission spectrum shows fringes vs. wavelength; a phase error shifts the fringes sideways, and the shift is read off directly.
- **Wafer-scale automatic testing:** many nominally identical MZIs across a wafer are measured by a probe station; histograms of the measured phase error, imbalance and insertion loss are the real-world version of our Monte Carlo distributions.

These are exactly the data that would test our predictions — and the data that would fix σ_β and σ_κ for a real process. Cite the source papers shown with each image before reusing them.

5 Code and tests

New in `topq/montecarlo.py`: `run_propagation_ensemble` (all lengths from one diagonalisation), `clean_amplitudes`, `phase_stats` (circular statistics). Scripts: `day6_mc_phase.py` (84 s), `day6_two_arm_schematic.py`. 3 new tests (agreement with direct propagation to 10^{-12}; T differential phase exactly zero under coupling disorder; circular-statistics limits). Suite: **96 passed**.

6 Self-test

1. Why does a chip-wide thickness error not appear in $\Delta\varphi$?
2. Why use the circular std instead of the ordinary std?
3. Show that the average fringe contrast left is $R = |\langle e^{i\Delta\varphi}\rangle|$.
4. Why does the phase error grow linearly with length?
5. Why does T's output *power* vary more than D's under gap errors, even though its phase does not?
6. What one measurement on a real wafer would tell you whether TOP-Q's protection is useful?